M201103W- Signals and Systems

ssch7_7 –Analysis and Characterization of LTI Systems Using the z-transform

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

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Analysis and Characterization of LTI Systems Using the z-transform

- System function and characterization of LTI systems
- System function and frequency response
- System function and the stability

1. System function and system characteristics in the z-domain

※ $H(z)$ and the zero-pole plot

$$H(z)=\frac{N(z)}{D(z)}=K\frac{(z-z_1)(z-z_2)\ldots(z-z_m)}{(z-p_1)(z-p_2)\ldots(z-p_n)}$$

Zero-pole gain form

$z_1, z_2, \ldots, z_m$ are zeros of system, using o to mark in the z plane.

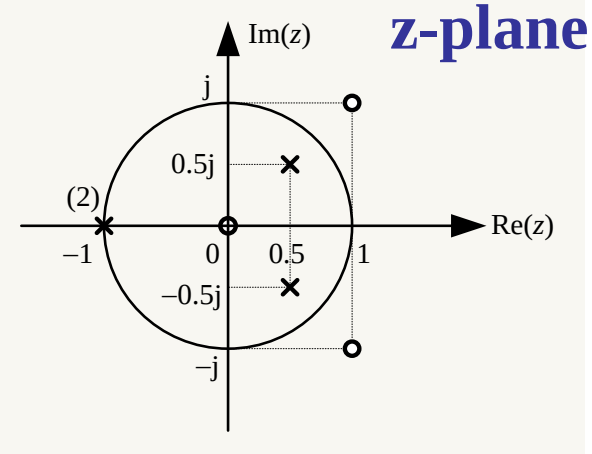
$p_1, p_2, \ldots, p_m$ are poles of system, using × to mark in the z plane.

Example. Draw zero-pole plot on the z-plane.

$$H(z)=\frac{z(z-1-\mathrm{j})(z-1+\mathrm{j})}{(z+1)^2(z-0.5-\mathrm{j}0.5)(z-0.5+\mathrm{j}0.5)}$$

Zeros: $z_1=0,\ z_2=1+\mathrm{j},\ z_3=1-\mathrm{j}$

poles: $p_4=0.5+0.5\mathrm{j},\ p_5=0.5-0.5\mathrm{j},\ p_{6,7}=-1$



1. System function and system characteristics in the z- domain

- Inverse z-transform of $H(z)$ is $h[k]$:

$$H(z)=\frac{N(z)}{D(z)}=K\frac{(z-z_1)(z-z_2)\ldots(z-z_m)}{(z-p_1)(z-p_2)\ldots(z-p_n)}=\sum_{i=1}^{n}\frac{k_i z}{z-p_i}$$

$$h[k]=\mathscr{Z}^{-1}\{H(z)\}=\sum_{i=1}^{n}k_i(p_i)^k u[k]$$

System characteristics of D-T LTI is determined by poles distribution of $H(z)$.

1. System function and system characteristics in the time domain

- Zero/pole distribution of $H(z)$ and system characteristics

(1) First order pole: $p=r$

$$h[k]=(r)^k u[k]$$

(2) conjugate poles

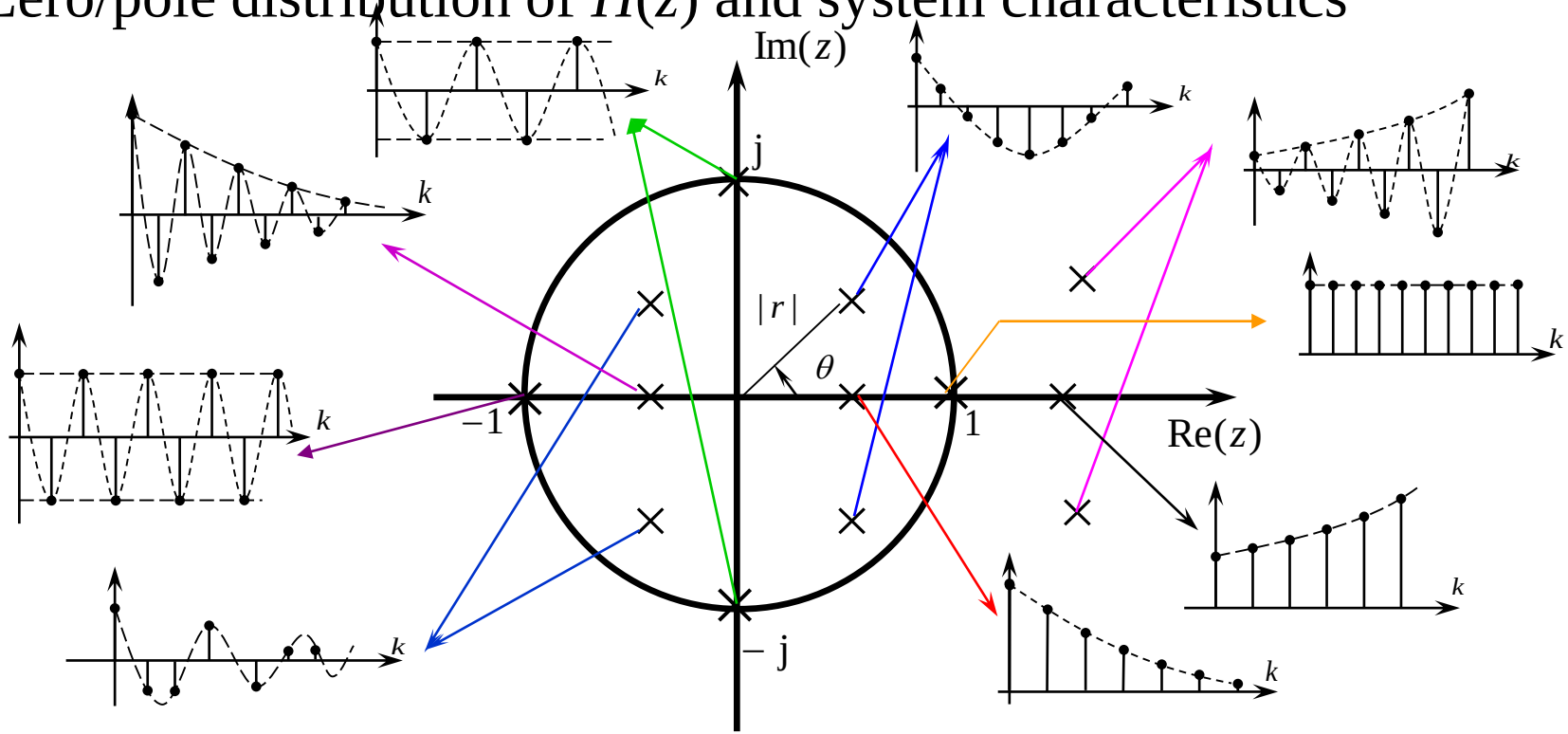
$$p_1=r\mathrm{e}^{\mathrm{j}\theta},\quad p_2=r\mathrm{e}^{-\mathrm{j}\theta}$$

$$H(z)=\frac{A}{1-r\mathrm{e}^{\mathrm{j}\theta}z^{-1}}+\frac{A^*}{1-r\mathrm{e}^{-\mathrm{j}\theta}z^{-1}}$$

Let A=1 $\quad h[k]=2(r)^k\cos(\theta\, k)u[k]$

1. System function and system characteristics in the time domain

- Zero/pole distribution of $H(z)$ and system characteristics



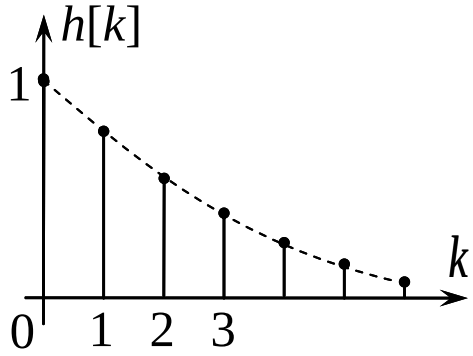
Example. The difference equation is $y[k] - a_1 y[k-1] = x[k]$. Analyse what the range of a_1 is to keep the causal system stable. Calculate $H(e^{j\Omega})$ and $h[k]$.

Solution: $$y[k] - a_1 y[k-1] = x[k]$$

System function: $$H(z) = \frac{z}{z - a_1} \quad \text{ROC: } |z| > |a_1|$$

To keep the system stable,
ROC should include the unit circle, that is $|a_1| < 1$.

$$h[k] = a_1^k u[k]$$



3. System function and stability

- Zero-pole distribution of $H(z)$ and stability

Theorem:

The sufficient and necessary condition to make system to keep a D-T LTI system stable is:

$$\sum_{k=-\infty}^{+\infty} |h[k]| < \infty$$

We can determine whether a system is stable through $H(z)$.

ROC of $H(z)$ including unit circle means the system is stable, where for a causal system, all poles are in the unit circle.

Relation between z-transform and s-transform

The mapping between z-plane and s-plane: $z = e^{sT}$

$$s = \sigma + j\omega \longrightarrow z = r \cdot e^{j\Omega}$$

$$z = e^{sT} = e^{(\sigma + j\omega)T}$$

$$= e^{\sigma T} \cdot e^{j\omega T}$$

$$= r \cdot e^{j\Omega}$$

Therefore: $r = e^{\sigma T}$

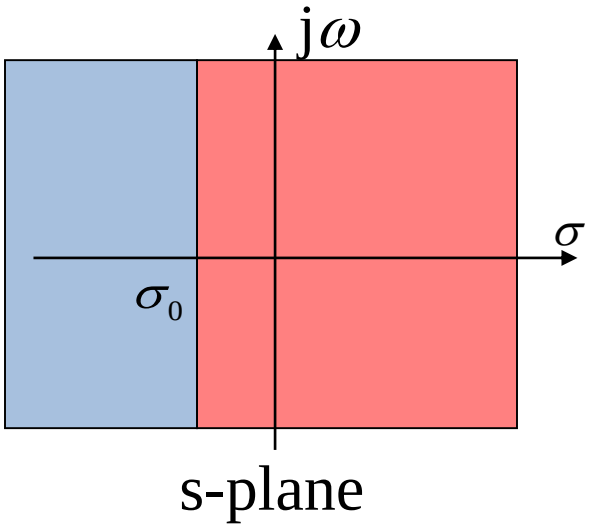
$\Omega = \omega T$

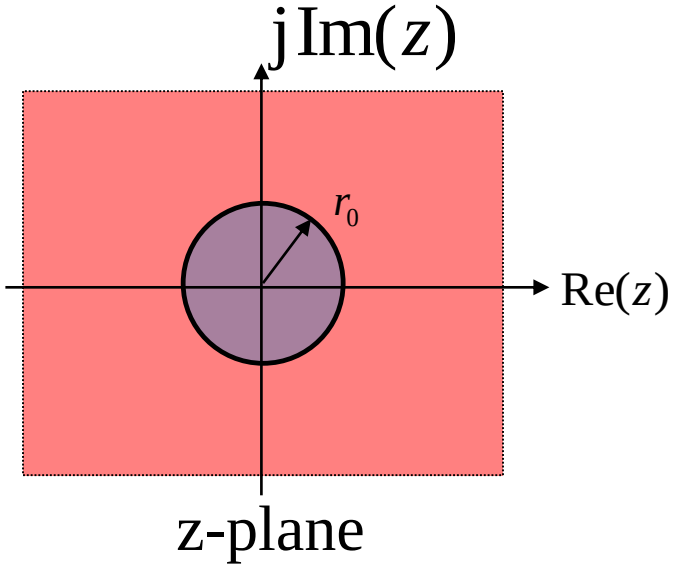
Relation between z-transform and s-transform

$$r = e^{\sigma T} \quad \Omega = \omega T$$

$$\sigma: \ \sigma < 0, \ \sigma > 0$$

$$r: \ r < 1, \ r > 1$$



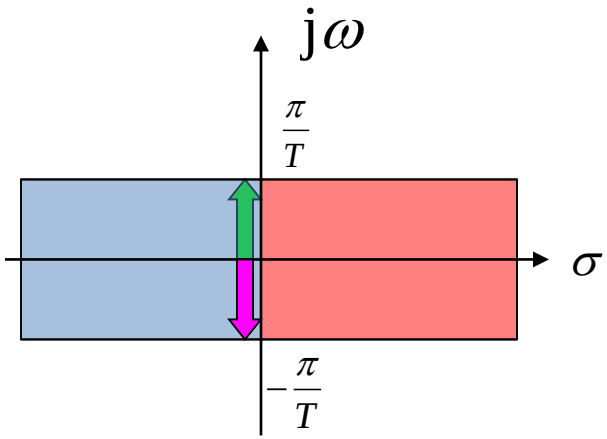


Relation between z-transform and s-transform

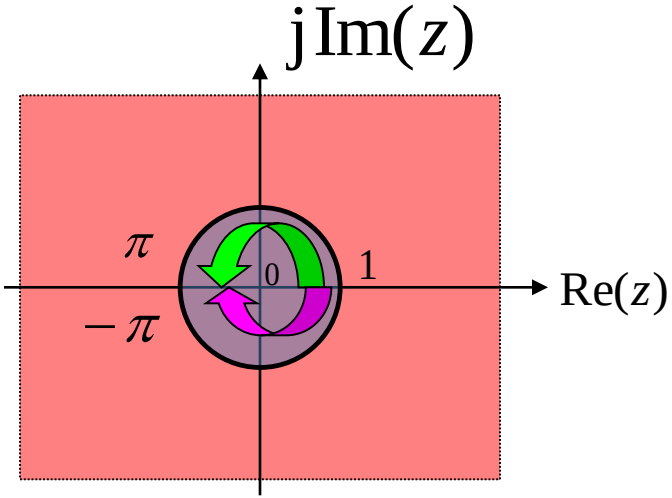
$$r = e^{\sigma T} \qquad \Omega = \omega T$$

$$\omega : 0 \to \frac{\pi}{T}, 0 \to -\frac{\pi}{T}$$

$$\Omega : 0 \to \pi, 0 \to -\pi$$



s-plane



z-plane

Example. Determine whether the system is stable and causal where $H(z) = \dfrac{1}{(1-0.5z^{-1})(1-1.5z^{-1})}$

Solution:

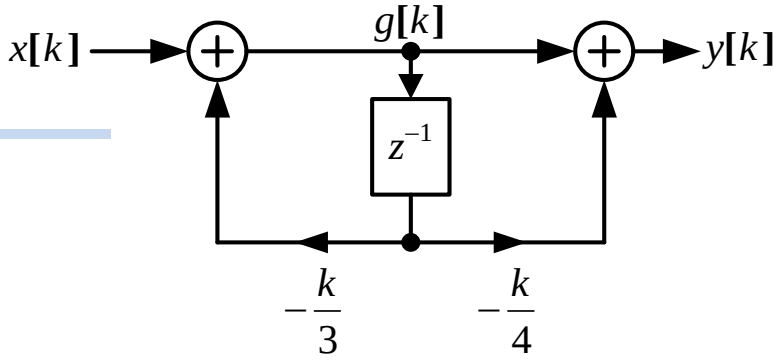
(1) $|z| < 0.5$ The system is neither stable nor causal.

(2) $0.5 < |z| < 1.5$ The system is stable, but not causal.

(3) $|z| > 1.5$ The system is not stable, but causal.

Obviously, there is no link between stability and causality.

Example. The diagram of a D-T causal system is shown as the figure. Calculate (a)$H(z)$. (b)Determine the range of k to keep the system stable.



$$G(z) = z^{-1}(-k/3)G(z) + X(z)$$

$$Y(z) = G(z) + (-k/4)z^{-1}G(z)$$

$$H(z) = \frac{Y(z)}{X(z)} = \frac{1-(k/4)z^{-1}}{1+(k/3)z^{-1}}$$

If the system is stable, $|k| < 3$